Integrating ICESat-2 and Sentinel-3 Data for Comprehensive K_{lidar} Retrieval: Case-I Water

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***Abstract*— The continuous demand for coastal zone resource exploitation and marine ecosystem protection challenges satellite remote sensing observation of shallow sea apparent optical properties. The existing satellite retrieval method of the light detection and ranging (LiDAR) attenuation coefficient (K_{lidar}) relies on water column photons, so it cannot achieve accurate water quality measurements in shallow water depths. In this study, the Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2) data were innovatively fused with Sentinel-3 image to retrieve comprehensive K_{lidar}. First, K_{lidar} is calculated in offshore areas using ICESat-2 water column photons and in nearshore areas using seafloor photons. ICESat-2-derived K_{lidar} was then integrated with Sentinel-3 image, employing Bayesian-optimized CatBoost (BO-CatBoost) to invert the spatial distribution of both offshore and nearshore areas. Finally, the retrieval results' weights were assigned based on Sentinel-3's band ratio, yielding comprehensive K_{lidar} results for the shallow seas. The proposed method successfully obtained an accurate spatial distribution of K_{lidar} in Passu Keah, Culebra, and Marquesas Keys. The results show that the coefficient of determination (R^2), mean relative error (MRE), and mean absolute error (MAE) for the three study areas range from 0.7402 to 0.8557, 0.0405 to 0.0885 m^{-1}, and 0.0033 to 0.0055 m^{-1}, respectively. The root mean square error (RMSE) ranges from 0.0043 to 0.0077 m^{-1}. It has higher reliability than Sentinel-3's $K_d(532)$. As satellite remote sensing improves, the method will efficiently map the LiDAR attenuation coefficient across vast areas, offering reliable data for resource development, water quality, and ecosystem conservation.**

***Index Terms*— Light detection and ranging (LiDAR) diffuse attenuation coefficient (K_{lidar}), Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2), Sentinel-3, shallow sea, CatBoost, Case-I water.**

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I. Introduction

SHALLOW sea, connected to coasts and islands, is a habitat for marine life and a frequent site for human activities [1]. Monitoring shallow sea's apparent quality is crucial for maintaining marine ecosystem health and promoting coastal economic sustainability [2]. In this field, the apparent optical properties of the water body are important indices for observation, which are related to the composition of the water body (such as suspended matter and phytoplankton) and vary with the external light field. Examples include the diffuse attenuation coefficient (K_d) and the light detection and ranging (LiDAR) attenuation coefficient (K_{lidar}), which can quantify the attenuation of radiative transmission in water [3], [4]. They directly affect the distribution of the underwater light field, the depth of the euphotic zone, and the photosynthetic efficiency of phytoplankton. Also, they play important roles in assessing water environmental conditions, underwater military applications, and underwater terrain mapping [5], [6], [7]. However, as water quality monitoring, management, and utilization become more complex, traditional in situ measurements struggle to meet the requirements of large scale and efficient monitoring [8].

Therefore, the ocean color remote sensing technology has become an effective supplement to traditional methods, leveraging its broad coverage and continuous data collection [9]. Remote sensing images capture water-leaving radiance (L_W) at various wavelengths, enabling the K_d retrieval over larger areas with appropriate models [10]. Semiempirical models like the quantitative algorithm for absorption (QAA) [11] combine radiative transfer models with empirical parameters to estimate coefficients like $K_d(490)$ from normalized L_W (nL_W). However, QAA's reliance on empirical parameters makes it sensitive to water turbidity, seafloor sediment, and sensor noise, limiting its effectiveness in complex shallow water [12], [13]. In comparison, empirical models like the blue–green band ratio [14] and chlorophyll concentration methods [15] are based on statistical links between multispectral radiance and K_d [16]. Ignoring the physics of light decay in water, however, these models also rely on the prior data, limiting their broad application in shallow seas.

Many researchers have proposed using spaceborne LiDAR to directly measure laser attenuation in water, thereby reducing the reliance on the prior data collection [17]. It emits laser pulses that interact with water-borne particles, with some energy returning to the receiver and the rest decaying [17].

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The laser energy loss is mainly caused by water absorption and scattering, which can be calculated by analyzing the echo signal and modeling the laser propagation path, referred to as K_{lidar} [18]. However, the traditional linear LiDAR detection system may not be able to capture discrete single photon events effectively when the echo signal is weak, resulting in incomplete waveform acquisition and reduced data acquisition efficiency. In order to meet this challenge and meet the need for high detection sensitivity and long detection range, photon counting LiDAR came into being [19].

The new LiDAR satellite, Ice, Cloud, and Land Elevation Satellite-2 (ICESat-2), equipped with the Advanced Topographic Laser Altimeter System (ATLAS). ATLAS's 532-nm blue–green laser penetrates water effectively [20], with a sensitive detector recording photons reflected or scattered by the water surface and column. Researchers use these water column photons to calculate K_{lidar}. Lu et al. [21] simplified the LiDAR equation to model K_{lidar} from water depth and laser energy, fitting photon count variations with depth. This method was verified for ICESat-2 derived K_{lidar} in environments without solar interference [22]. Watkins et al. [23] confirmed ICESat-2's accuracy by comparing field measurements of lake depths and K_{lidar}. However, due to the inherent limitations of the data acquisition method, ICESat-2's K_{lidar} detection results are confined to the satellite's laser trajectory. As reported by Zhang et al. [24], K_{lidar} shows a strong correlation with image-based $K_d(532)$, suggesting that integrating multispectral images with ICESat-2 data can effectively overcome the spatial constraints imposed by the satellite's trajectory.

The integration of active and passive remote sensing data from satellite platforms shows significant potential in K_{lidar} retrieval for bathymetry and seafloor sediment classification. Chen et al. [25] retrieved K_{lidar} for China's Xiaolangdi Reservoir using ICESat-2 water column photons and Sentinel-3 image, achieving higher accuracy than image-based products. Zhang et al. [24] combined ICESat-2 derived K_{lidar} with Sentinel-2 image and applied the Stumpf empirical model to retrieve water depth information for Zhongye Island and Reefs, Guam, and Ngau, with a maximum mean absolute error (MAE) of 1.52 m and root mean square error (RMSE) of 1.89 m. Zhu et al. [26] proposed an IDBM model based on the QAA [11] algorithm, which, after obtaining K_{lidar}, used the Jerlov spectral curve and sensor spectral response function to derive K_d from Sentinel-2 blue–green bands and applied a dual-band model for water depth retrieval, achieving an average MAE and RMSE of 1.23 and 1.67 m, respectively. Wang et al. [27] developed a semianalytical model to retrieve bottom reflectance (R_b) under various substrates using LiDAR-observed attenuation coefficients and water depth values. However, these studies rely on the sufficient water column photons. As water depth decreases, the photon count may fall below the requirement for current K_{lidar} retrieval algorithms, leading to unreliable ICESat-2-derived K_{lidar} in nearshore areas. Furthermore, although nearshore waters are generally turbid, these studies are based on the assumption that the water is equally clean. Thus, the absence of an ICESat-2 K_{lidar} calculation method for nearshore waters will inevitably introduce errors in the fused inversion outcomes.

In response to these limitations, this study proposes a comprehensive K_{lidar} retrieval method that integrates multisource satellite data. The innovation lies in three aspects. First, K_{lidar} is calculated using ICESat-2 water column and seafloor photons to accommodate varying water depths. Second, the K_{lidar} distribution in shallow sea is retrieved by combining these data with Sentinel-3 image and applying Bayes-optimized CatBoost [28]. Finally, the method segments nearshore and offshore areas based on Sentinel-3 spectral ratios, assigning different weights to integrate the retrieval results, thus enabling K_{lidar} mapping for the entire shallow sea area. This approach is expected to enhance the accuracy and efficiency of marine water quality monitoring and pave the way for satellite-based water quality retrieval.

II. Study Sites and Data

A. Study Sites and Airborne Data

As shown in Fig. 1, three study sites with Case-I water conditions were selected for this study. Passu Keah, as depicted in Fig. 1(a), was chosen as the first study area. It is located at a latitude of 16.02°N–16.05°N and longitude of 111.45°E–111.50°E, southwest of the Xisha Islands in South China sea. The island, a coral atoll, occupies approximately 0.4 km^2 and extends about 10 km from east to west. A low-lying bar, about 2.5 m above sea level, is presented on the western side of the atoll. The water in this area is clear, with depths ranging from 25 to 45 m. In June 2020, we utilized the Mapper5000 full-waveform LiDAR system on an aircraft to collect data from the Passu Keah shallow sea area at an altitude of approximately 500 m. The Mapper5000 has a laser repetition frequency of 5 kHz, a pulsewidth of 0.95 ns, and the vertical resolution of the obtained waveform data in water is 0.02 m [29]. As shown in Fig. 1(a), 210 shallow sea points were selected, and the waveform data at these points were processed using the waveform exponential fitting method proposed by Billard et al. [30], with the corresponding results serving as verification data.

Culebra in Puerto Rico and the Marquesas Keys in Florida were also selected for study. As shown in Fig. 1(b), Culebra is situated east of the main island of Puerto Rico at latitudes of 18.25°N–18.36°N and longitudes of 65.15°W–65.35°W, covering an area of approximately 22 km^2. The seafloor is primarily composed of sand and gravel. Fig. 1(c) shows that the Marquesas Keys, located in Southwest Florida, consist of shoals, lagoons, and reefs, with a total area of approximately 29 km^2, including the lagoon. Both study areas with Case-I water conditions had clear water but lacked the K_{lidar} measured by airborne LiDAR. To address this, highly accurate underwater terrain data, converted from underwater topography measured by the National Oceanic and Atmospheric Administration (NOAA) using Airborne Laser Bathymetry (ALB), was introduced as a reference. As shown in Fig. 2, Culebra's ALB data were acquired between January 20, 2019 and June 2, 2019, with a bathymetry accuracy of 12.1 cm. The

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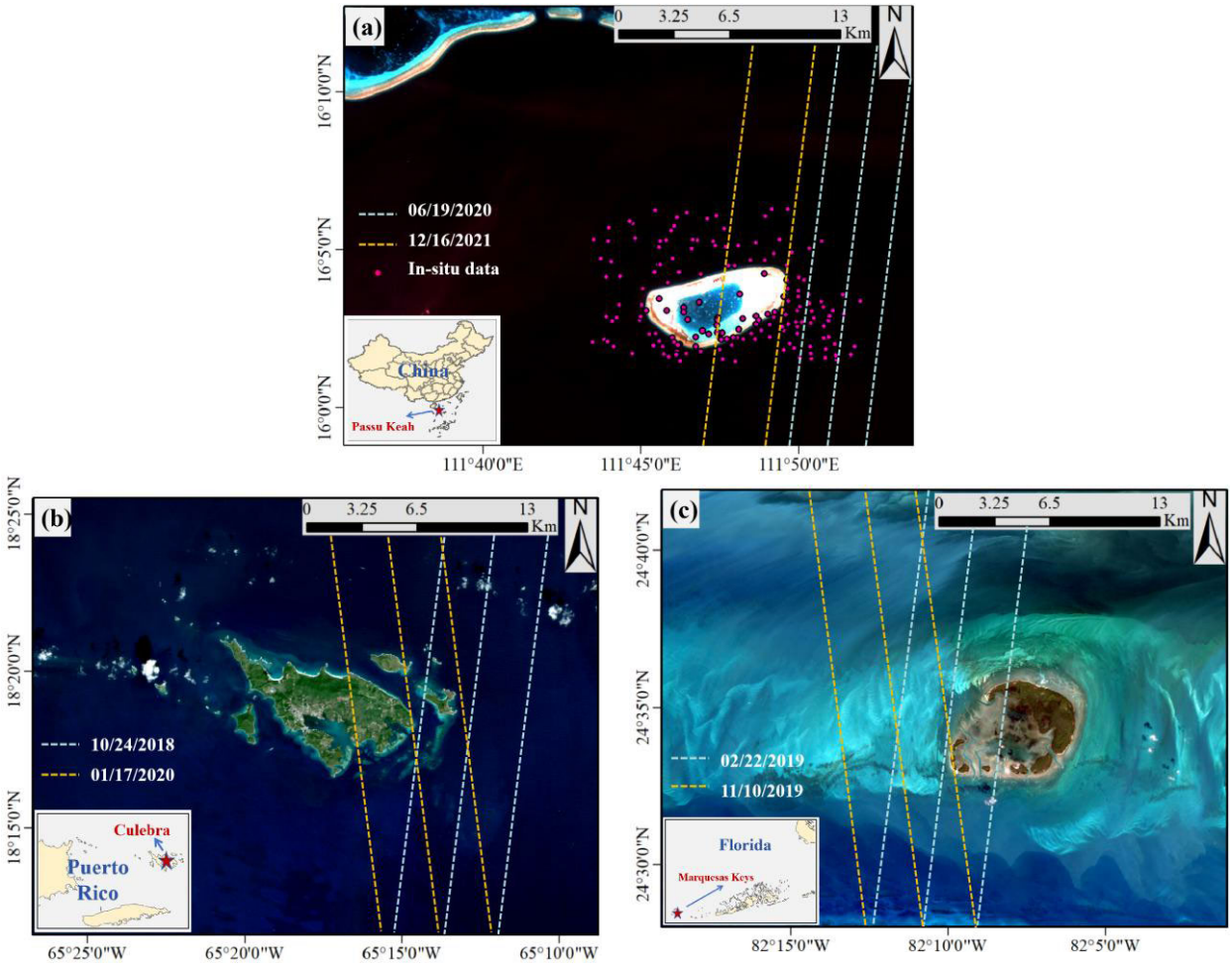


Fig. 1. Study site. (a) Passu Keah. (b) Culebra. (c) Marquesas Keys. The blue and yellow dashed lines correspond to ICESat-2 data from different dates, while the red dots in (a) represent the coordinates of the in situ K_{lidar}.

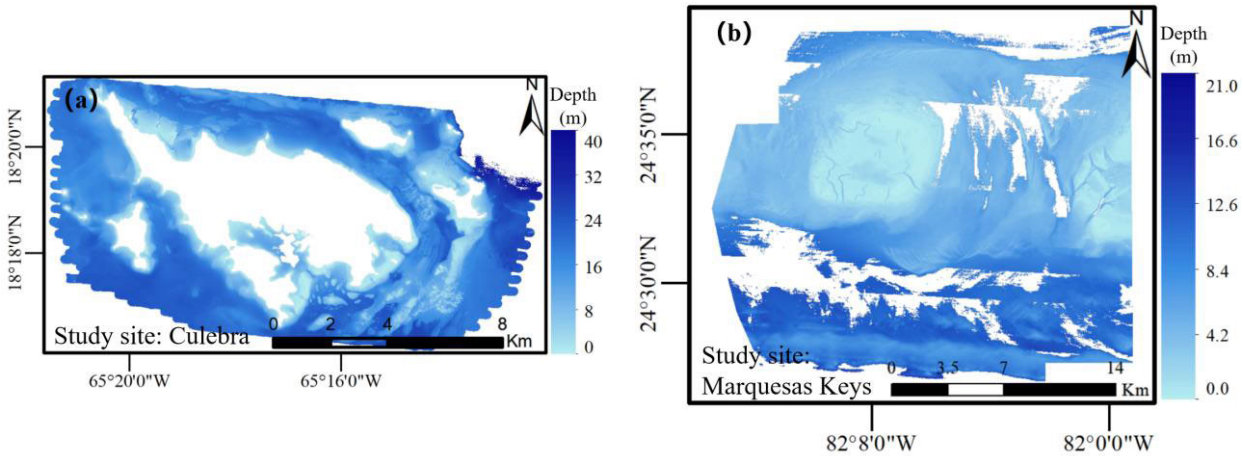


Fig. 2. Airborne bathymetry results. (a) Culebra. (b) Marquesas Keys. Land and areas with missing data are represented in white.

Marquesas Keys' ALB data were acquired between November 20, 2018 and March 23, 2019, with a bathymetry accuracy of 15.1 cm.

B. ICESat-2 ATL03 Data

ICESat-2's ATLAS operates in a photon-counting mode, which significantly enhances detection sensitivity and reduces the laser energy requirement. Photon-counting mode allows the laser to function at high repetition frequencies. ATLAS's laser has a frequency of 10 kHz and emits a blue–green beam at a wavelength of 532 nm. Laser at this wavelength can penetrate the water surface, where it is absorbed, scattered, and reflected by the water column. Some of these laser pulses can reach the ocean floor. The apparent optical properties

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TABLE I
DETAILED INFORMATION ON THE STUDY SITES AND CORRESPONDING DATA

Data	Passu Keah	Culebra	Marquesas Keys
ICESat-2 ATL03 data	06/19/2020 (Deep) 12/16/2021 (Deep and Shallow)	10/24/2018 (Deep) 01/17/2020 (Deep and Shallow)	02/22/2019 (Deep and Shallow) 11/10/2019 (Deep)
Sentinel-3 OL_2_WFR image	06/20/2020 12/15/2021	10/24/2018 01/19/2020	02/24/2019 11/19/2019
Airborne data	K_{lidar}	Underwater topography	Underwater topography

of the water influence the ability of ICESat-2 to penetrate water. By analyzing the number of photons from both the water column and the seafloor, it is possible to extract K_{lidar} from the raw photon data. The ICESat-2 ATL03 data record the time, longitude, latitude, and elevation of these photons. To minimize the interference from solar background noise, this study utilized nighttime data. Table I lists the selected ICESat-2 data information and outlines the scope of ICESat-2's applicability.

In this study, sea surface photons, water column photons, and seafloor photons from the ICESat-2 ATL03 data require minimal processing for extraction. Initially, an elevation histogram divides the positions into sea surface, water column, and seafloor categories, allowing for the extraction of sea surface and water column photons within their respective ranges [31]. Next, prepruning quadtree isolation [32], a classical noise removal algorithm, is applied to extract seabed photons from the remaining data. Finally, the photon parameter tracking method [33] corrects for the refraction of water column and seafloor photons, eliminating refraction errors caused by the laser pulses at the water–air interface.

C. Sentinel-3 OL_2_WFR Image

Sentinel-3's Ocean and Land Color Instrument (OLCI) provides medium-resolution optical images of shallow ocean areas, with repeat observations in less than 2 days. This frequency provides a sufficient number of multispectral images for data inversion. The OLCI level-2 water products offer water and atmospheric geophysical parameters at full or reduced resolution, named OL_2_WFR and OL_2_WRR, respectively. The OL_2_WFR product has a spatial resolution of 300 m and includes 21 bands labeled from Oa1 to Oa21. As shown in Table I, Sentinel-3 OL_2_WFR images were selected for two periods in each study area. They were downloaded free of charge from the Sentinel Scientific Data Hub of the European Space Agency (ESA). Sentinel-3 images were chosen to closely match the acquisition time of the ICESat-2 data, with the time interval between the two datasets not exceeding 9 days. The $K_d(490)$ product provided by Sentinel-3 was calculated by Morel et al. [15] and can be downloaded simultaneously with the image data.

III. METHODS

As shown in Fig. 3, the study of retrieving shallow sea K_{lidar} using ICESat-2 and Sentinel-3 data involves three steps. First, ICESat-2 $K_{\text{lidar}}^{\text{off}}$ is calculated using water column photons in the offshore region based on the LiDAR equation. $K_{\text{lidar}}^{\text{near}}$ is calculated by combining the water depth from seafloor photons with the corresponding Sentinel-3 blue–green band ratio in the nearshore region. Second, the ICESat-2-derived K_{lidar} values from different regions are matched with the pixel values of Sentinel-3 multispectral images. The Bayes-optimized CatBoost (BO-CatBoost) is then used to retrieve the K_{lidar} results for both the offshore and nearshore regions, respectively. Third, Sentinel-3 spectral combinations are used to calculate the weights for the results from the offshore and nearshore regions. These two results are then fused according to their weights to map the spatial distribution of K_{lidar} in the shallow sea fully.

A. Offshore K_{lidar} Calculation Based on Water Column Photons

It has been proposed that the calculation method for K_{lidar} requires a specific water depth in the shallow sea to ensure a sufficient number of water column photons and to minimize the influence of seafloor terrain fluctuations on optical properties [21]. As shown in the following equation, the LiDAR diffuse attenuation coefficient $K_{\text{lidar}}^{\text{off}}$ can be effectively estimated at this time based on the relationship between the number of photons in the water column and the water depth:

$$S(z) = A\beta(\pi, z)\, e^{-2K_{\text{lidar}}^{\text{off}} z} \tag{1}$$

where $S(z)$ represents the LiDAR signal at depth z, A is a determined LiDAR constant, $\beta(\pi, z)$ is the volume backscattering coefficient, and $K_{\text{lidar}}^{\text{off}}$ is a coefficient that lies between the beam attenuation coefficients c and K_d, depending on the size of the LiDAR spot on the ocean surface. For ICESat-2, $K_{\text{lidar}}^{\text{off}}$ can be approximated as $K_d(532)$.

Assuming that the water body is uniform, $K_{\text{lidar}}^{\text{off}}$ can be represented by the exponential attenuation of water column photons with respect to water depth. Specifically, the water column photons are segmented according to the distance along the satellite orbit. Within each segment, the water column photons are further divided into several bins based on elevation to accurately analyze the distribution and attenuation of photons in different water layers. In this study, the ICESat-2 data are divided into 0.001° along-track segments, and the water column photons are further divided into 0.1-m data bins. The number of water column photons in each layer is counted, and the probability density is calculated based on the elevation distribution. As shown in the following equation, the degree of water attenuation to the laser can be understood by fitting

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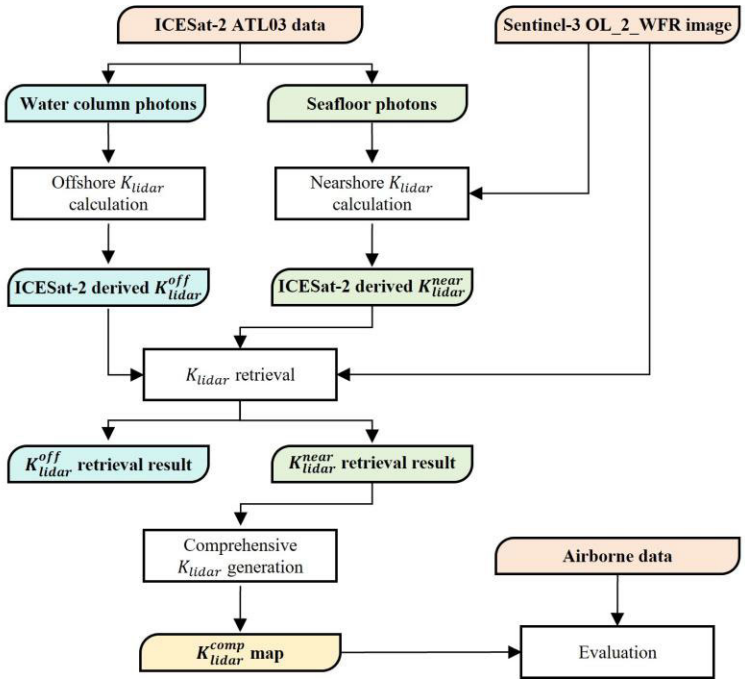


Fig. 3. Workflow.

the probability change of the water column photon with water depth using the index:

$$K_{\text{lidar}}^{\text{off}} = -\frac{1}{2}\frac{d}{dz}\ln\left(S\left(z\right)\right). \tag{2}$$

B. Nearshore K_{lidar} Calculation Based on Seafloor Photons

The number of water column photons decreases with the decrease of water depth, rendering the exponential fitting method based on water column photons unsuitable for the nearshore region. Therefore, a new calculation method for nearshore K_{lidar} is still needed. Morel et al. [15] discovered that the apparent optical properties of Case-I water are primarily influenced by floating matter in the water and its metabolic and degradation products. Building on this, Zhang et al. [24] proposed a Satellite Derived Bathymetry (SDB) method that does not require prior information. Initially, $K_{\text{lidar}}^{\text{off}}$ is calculated through ICESat-2 water column photons. Then, chlorophyll concentrations (Chla) are used as a medium to bind the ICESat-2 derived $K_{\text{lidar}}^{\text{off}}$ to the corresponding Sentinel-3 water depth conversion.

Inspired by the above research, Chla is used as a medium for converting water depth to K_{lidar} in nearshore areas, named $K_{\text{lidar}}^{\text{near}}$. In fact, ICESat-2's maximum bathymetry range in Case-I water can reach approximately 40 m, with an accuracy better than 1 m [34]. In the nearshore region, it can provide accurate water depth along the track direction through seafloor photons. Concurrently, the water depth z measured by ICESat-2 in SDB should satisfy the following equation [35]:

$$z = m_0\frac{\ln\left[n\cdot r_{\text{rs}}\left(\text{blue}\right)\right]}{\ln\left[n\cdot r_{\text{rs}}\left(\text{green}\right)\right]} - m_1 \tag{3}$$

where n is a fixed constant with a value of 1000 and m_0 and m_1 can be expressed in the exponential form of Chla in the following equations:

$$m_0 = 52.083\cdot e^{0.957\cdot\text{Chla}} \tag{4}$$

$$m_1 = 50.156\cdot e^{0.957\cdot\text{Chla}} \tag{5}$$

where the constants 52.083 and 50.156 are derived from the simulation model of pure water [36], while the constant 0.957 is an empirical setting of Sentinel-3 images [37].

r_{rs} is the subsurface remote sensing reflectance, which can be converted from Sentinel-3 image reflectance (R_{rs}) according to the following expression:

$$r_{\text{rs}}\left(\lambda\right) = \frac{R_{\text{rs}}\left(\lambda\right)}{0.52 + 1.7R_{\text{rs}}\left(\lambda\right)}. \tag{6}$$

Therefore, the relationship between Chla and water depth z can be expressed as follows:

$$\text{Chla} = \frac{\ln\left[z/\left(52.083L - 50.156\right)\right]}{0.957} \tag{7}$$

where L is the logarithmic ratio term in expression (3)

Meanwhile, $K_d(490)$ and Chla can be converted into one another [36]

$$K_d\left(490\right) = K_{dw}\left(490\right) + 0.069\cdot\text{Chla}^{0.702} \tag{8}$$

where $K_{dw}(490)$ represents the diffuse attenuation coefficient of pure water at 490 nm, with a value of 0.0217 m^{-1}. The conversion between $K_d(490)$ and $K_{\text{lidar}}^{\text{near}}$ is as follows:

$$K_{\text{lidar}}^{\text{near}} = 0.68\left(K_d\left(490\right) - 0.022\right) + 0.054. \tag{9}$$

From these equations, the combined expressions (7)–(9) can be used to determine $K_{\text{lidar}}^{\text{near}}$ in the nearshore area from the water depth z measured by ICESat-2.

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C. K_{lidar} Retrieval Based on BO-CatBoost

The K_{lidar} calculated values for water column photons and seafloor photons are limited to the ICESat-2 trajectory. By combining the K_{lidar} derived from ICESat-2 with Sentinel-3 R_{rs}, it is possible to extend the coverage of K_{lidar} from the ICESat-2 trajectory to the entire study area. Specifically, based on K_{lidar} estimation results, the Sentinel-3 image is spatially interpolated to obtain R_{rs}, and they are used as the learning sample of CatBoost to construct the complex nonlinear relationship between them. Based on these results, a wide range of R_{rs} is input to the trained model, achieved K_{lidar} retrieval in the same region. It is important to note that the O1–O6 bands of Sentinel-3 images are used in this study. Each of these bands has a signal-to-noise ratio (SNR) of over 1000. The high SNR not only ensures the reliability and accuracy of R_{rs} but also provides high-quality input data for K_{lidar} retrieval, and enables weak signals to be detected, reflecting subtle optical changes in the water body.

CatBoost [28] is an advanced machine learning model based on the gradient boosting algorithm, which iteratively trains a series of weak learners to improve its prediction performance gradually. It excels when dealing with complex datasets. The core optimization process of CatBoost relies on minimizing the objective function $F(y, f)$

$$F(y, f) = \sum_{i=1}^{n} \ell\left(y_i, f(x_i)\right) + \sum_{j=1}^{m} \Omega\left(f_j\right) \tag{10}$$

where $\ell(y_i, f(x_i))$ represents the loss function, measuring the error between the model's predicted value $f(x_i)$ and the true value y_i. The common loss functions include exponential loss. $\Omega(f_j)$ is a regularization coefficient that controls the model's complexity to prevent overfitting and is often related to the complexity of the decision tree.

CatBoost's performance in K_{lidar} inversion is highly dependent on hyperparameter-like learning rate, number, and tree depth. These hyperparameters significantly influence the model's ability to capture the nonlinear relationship between K_{lidar} and R_{rs}. Therefore, Bayesian optimization (BO) is employed for hyperparameter, obtained BO-CatBoost. This approach uses the global search capabilities of BO to find the optimal hyperparameter set, enhancing the model's retrieval performance. The method utilizes a Gaussian process (GP) as a surrogate model to guide the hyperparameter search. The GP, a nonparametric Bayes technique, is defined by a mean function $m(x)$ and a variance function $k(x, x')$

$$f(x) \sim \mathrm{GP}\left(m(x), k\left(x, x'\right)\right). \tag{11}$$

On the other hand, BO-CatBoost employs expected improvement (EI) to determine the following search location:

$$\mathrm{EI}(x) = E\left[\max\left(0, y_{\text{best}} - f(x)\right)\right] \tag{12}$$

where y_{best} represents the best objective function value observed so far. BO-CatBoost identifies the param point x that maximizes the EI value, effectively selecting the optimal hyperparameter combination for evaluation and optimization. This approach ensures accurate and efficient K_{lidar} retrieval.

D. Comprehensive K_{lidar} Generation Based on Weighted Fusion

The ICESat-2 K_{lidar} calculation methods were adapted to account for variations in water turbidity and photon distribution characteristics between the offshore and nearshore of the shallow sea. BO-CatBoost was then applied for retrieval. Therefore, a weighted fusion technique was employed to integrate the retrieval results and obtain a comprehensive K_{lidar} result [38].

The shallow sea environment is rapidly changing and complex, making it unrealistic to accurately define the boundary between the off and nearshore areas. To obtain a smooth and comprehensive K_{lidar} result, the final diffuse attenuation coefficient $K_{\text{lidar}}^{\text{Comp}}$ should be calculated using the following equation:

$$K_{\text{lidar}}^{\text{Comp}} = (1 - W)\, K_{\text{lidar}}^{\text{off}} + W K_{\text{lidar}}^{\text{near}} \tag{13}$$

where weight W is based on the following equation:

$$W = \begin{cases} 0, & R \le 0.2604 \\ -1.175 + 4.512R, & 0.2604 < R < 0.4821 \\ 1, & R \ge 0.4821 \end{cases} \tag{14}$$

where R is the ratio of Sentinel-3 R_{rs} at 665 nm [$R_{\text{rs}}(665)$] and 490 nm [$R_{\text{rs}}(490)$] wavelengths

$$R = R_{\text{rs}}(665) / R_{\text{rs}}(490). \tag{15}$$

Thus, (13) ensures a smooth transition between the two K_{lidar} results. When R is small, $K_{\text{lidar}}^{\text{Comp}}$ is primarily composed of $K_{\text{lidar}}^{\text{off}}$ in the offshore region. Conversely, in the nearshore region, $K_{\text{lidar}}^{\text{Comp}}$ is mainly composed of $K_{\text{lidar}}^{\text{near}}$.

E. Evaluation

Airborne K_{lidar} data or water depth data are used to evaluate the accuracy of the K_{lidar} results across different study areas. At Passu Keah, the airborne K_{lidar} data collected in June 2020 were directly compared with the retrieval K_{lidar} results. In Culebra and Marquesas Keys, where corresponding airborne K_{lidar} data were unavailable, this study conducted random and uniform sampling of high-precision water depth data collected by the airborne LiDAR System (ALS) during the same period. The extracted water depth points were converted into K_{lidar} using the method proposed by Zhang et al. [24], denoted by $K_{\text{lidar}-z}$, and compared with the retrieval K_{lidar} results.

MAE, mean relative error (MRE), coefficient of determination (R^2), RMSE, and slope were used as evaluation indices for retrieval accuracy, as defined in the following equations (16)–(20):

$$\mathrm{MAE} = \frac{1}{n} \sum_{i=1}^{n} \left| y_i - \hat{y}_i \right| \tag{16}$$

$$\mathrm{MRE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{17}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2} \tag{18}$$

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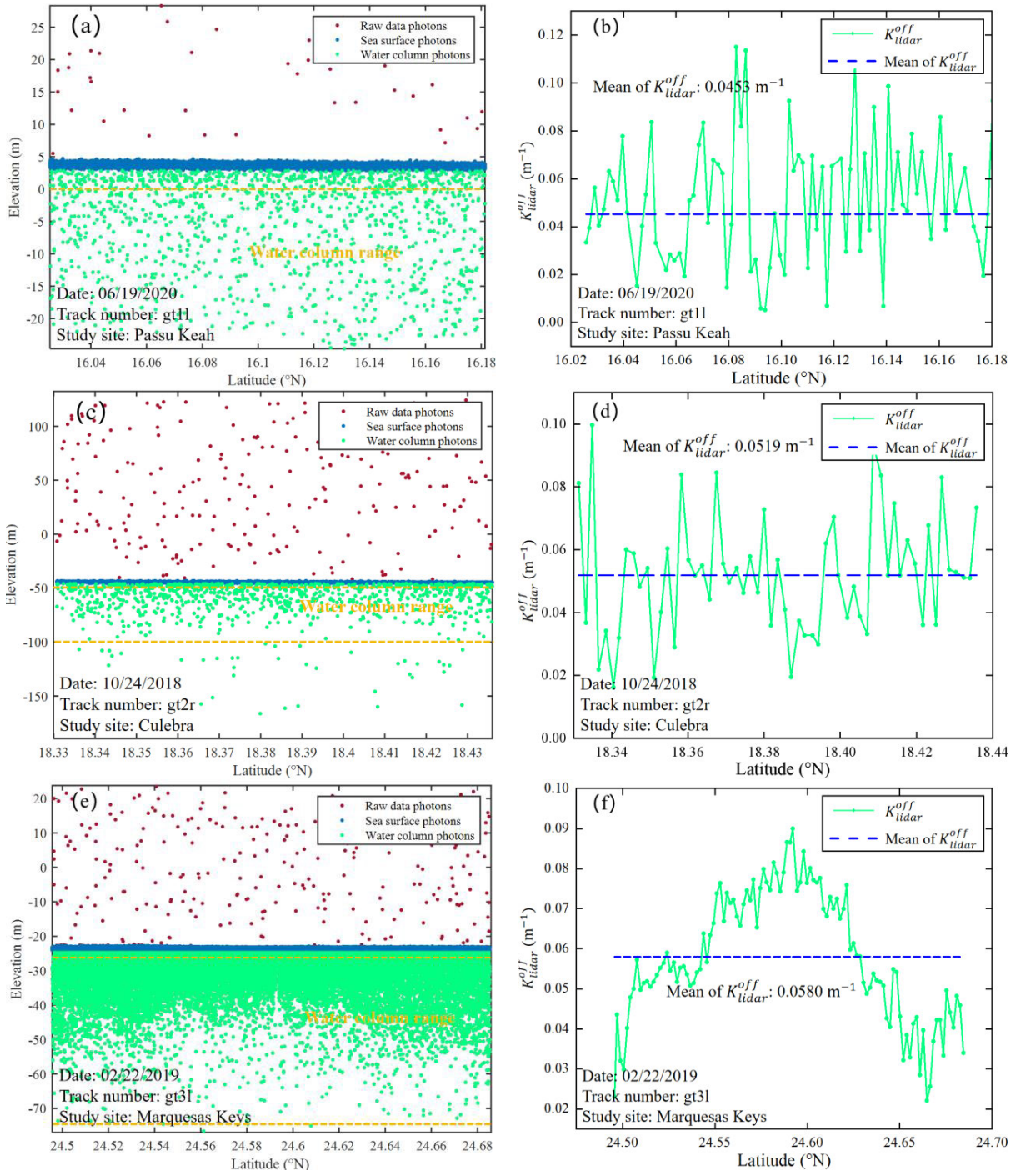


Fig. 4. ICESat-2 water column photon extraction and corresponding $K_{\text{lidar}}^{\text{off}}$ results. (a), (c), and (e) Photon extraction results from the ICESat-2 water column at offshore region of Passu Keah, Culebra, and Marquesas Keys, where red dots represent the raw noise photons, blue dots represent the surface photons, and green dots represent the water column photons. (b), (d), and (f) Corresponding $K_{\text{lidar}}^{\text{off}}$ results, where the green line shows the $K_{\text{lidar}}^{\text{off}}$ value and the blue line shows the mean $K_{\text{lidar}}^{\text{off}}$ value.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{19}$$

$$\text{Slope} = \frac{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)\left(\hat{y}_i - \bar{\hat{y}}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)^2} \tag{20}$$

where n represents the number of samples, y_i denotes the value of the airborne data, $\hat{y}_i$ indicates the result of the satellite retrieval, and $\bar{y}$ signifies the average value of the airborne data.

IV. Results

A. Evaluation of ICESat-2 Derived K_{lidar} Results

In this section, the ICESat-2-derived K_{lidar} results for both offshore and nearshore areas were qualitatively evaluated. Fig. 4 shows the spatial distribution of photons for Passu Keah, Culebra, and Marquesas Keys in the offshore region, along with their corresponding $K_{\text{lidar}}^{\text{off}}$ from offshore. Fig. 5 displays $K_{\text{lidar}}^{\text{near}}$ from nearshore for these study sites.

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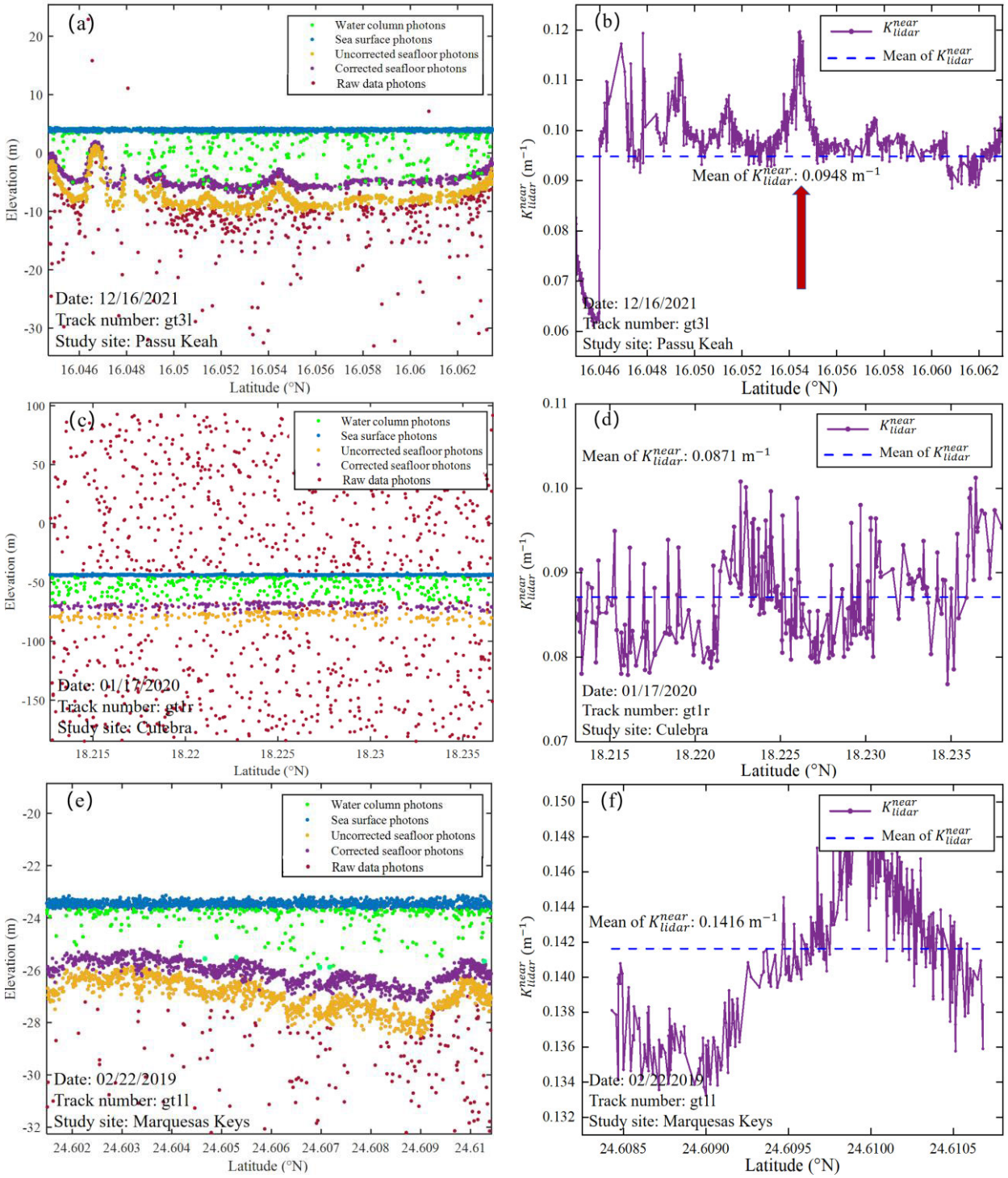


Fig. 5. ICESat-2 water column photon extraction and corresponding $K_{\text{lidar}}^{\text{near}}$ results. (a), (c), and (e) Photon extraction results from the nearshore region, where yellow and purple dots represent the uncorrected and corrected seafloor photons, respectively. (b), (d), and (f) Corresponding $K_{\text{lidar}}^{\text{near}}$ results, where the purple line shows the $K_{\text{lidar}}^{\text{near}}$ value and the blue line shows the mean $K_{\text{lidar}}^{\text{near}}$ value.

Fig. 4(a), (c), and (e) displays the water column photon extraction results from ICESat-2 in the offshore regions of Passu Keah, Culebra, and Marquesas Keys, respectively. The results indicate that ICESat-2 cannot detect significant seafloor reflection signals. However, due to its intense water body penetration, it captured a sufficient number of photon signals within the water column. Notably, the number of water column photons decreases as water depth increases, offering a basis for assessing the laser beam's attenuation in the water body.

Using water column photons at depths of 3–50 m below the sea surface, we calculated $K_{\text{lidar}}^{\text{off}}$. Fig. 4(b), (d), and (f) presents the $K_{\text{lidar}}^{\text{off}}$ results for ICESat-2 data in Passu Keah, Culebra, and Marquesas Keys, respectively. Passu Keah has an average $K_{\text{lidar}}^{\text{off}}$ of 0.0453 m^{-1}, Culebra's is 0.0519 m^{-1}, and Marquesas

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Keys' is 0.0580 m^{-1}. The results indicate that Passu Keah has the most transparent water, while Marquesas Keys is the most turbid. Further analysis revealed that the Marquesas Keys' seafloor sediment, mainly composed of coral reefs and limestone, supports rich underwater biological activity. The water body there contains a high concentration of dissolved and suspended substances, which enhances the laser beam's attenuation effect.

Fig. 5(a), (c), and (e) depicts the spatial distribution of ICESat-2 photons in the nearshore regions of Passu Keah, Culebra, and Marquesas Keys, respectively. It is observed that the nearshore region has a limited number of water column photons, restricting the calculation of K_{lidar} based on these photons. However, the intense penetration of ICESat-2's laser pulse through water allows for the use of sufficient seafloor photons for K_{lidar} calculations in the nearshore area. The refraction correction is applied to eliminate errors caused by the laser pulse's passage through the water–air interface. In the figures, yellow dots represent seafloor photons before refraction correction, while purple dots indicate those after correction.

Refraction-corrected seafloor photons are used to calculate K_{lidar}^{near} in the nearshore region. Fig. 5(b), (d), and (f) displays the K_{lidar}^{near} calculation results from ICESat-2 for Passu Keah, Culebra, and Marquesas Keys, respectively. The results indicate that the optical properties of the nearshore water body are significantly influenced by seafloor terrain. High K_{lidar}^{near} values, indicating turbid water with low transparency, are found in areas with steep seafloor terrain and shallow water depths due to increased reflection from bottom materials and suspended matter. Conversely, low K_{lidar}^{near} values, suggesting more transparent water and higher transparency, are observed in areas with flat seafloor terrain and more excellent water depths. At the location marked by the red arrows in Fig. 5(a) and (b) (16.054°N), the rugged terrain, decreased water depth and increased K_{lidar}^{near} to 0.12 m^{-1} are noted. Passu Keah's nearshore area, with its varied submarine terrain, has an average K_{lidar}^{near} of 0.0948 m^{-1}. In contrast, Culebra's nearshore terrain, which is relatively stable, has an average K_{lidar}^{near} of 0.0871 m^{-1}. The nearshore area of Marquesas Keys, as shown in Fig. 5(e) and (f), is predominantly shallow (2–3 m), leading to higher K_{lidar}^{near} values. The region demonstrates high scattering and strong absorption properties in response to the ICESat-2 laser pulse.

B. Evaluation of Comprehensive K_{lidar} Results

As shown in Fig. 6, comprehensive K_{lidar} results for Passu Keah, Culebra, and Marquesas Keys are presented, including K_{lidar}^{Comp} distribution map from two different dates for each study area. Generally, K_{lidar}^{Comp} values are higher in the nearshore region and lower in the offshore region. By comparing the water depth distributions in Fig. 2(a) with the corresponding K_{lidar}^{Comp} variations in Fig. 6(c) and (d) or Fig. 2(b) with Fig. 6(e) and (f), it can be qualitatively concluded that there is a negative correlation between water depth and K_{lidar}^{Comp}. This phenomenon is likely due to the fact that as water depth decreases, bottom materials, such as sand and gravel, are more susceptible to wave action, increasing the turbidity of the shallow waters.

TABLE II
K_{lidar}^{Comp} MEAN VALUE AND SD

Study Site	Date	Mean value (m^{-1})	SD (m^{-1})
Passu Keah	06/20/2020	0.0567	0.0202
	12/15/2021	0.0509	0.0222
Culebra	10/24/2018	0.0614	0.0053
	01/19/2020	0.0671	0.0074
Marquesas Keys	02/24/2019	0.1029	0.0265
	11/19/2019	0.0962	0.0172

The K_{lidar}^{Comp} results from the two periods within each study area were further compared. Table II corresponds to Fig. 6 and presents the mean of these K_{lidar}^{Comp} values. Despite variations in K_{lidar}^{Comp} retrieval across different periods, the overall trend of change is highly consistent. This consistency indicates that the water quality in Passu Keah, Culebra, and Marquesas Keys was similar during the study's data collection period.

The mean K_{lidar}^{Comp} values for Passu Keah were 0.0567 and 0.0509 m^{-1}, indicating the clearest water quality among the three study areas. Culebra's water was more turbid than Passu Keah's, and Marquesas Keys had the most turbid water. Table II also details the K_{lidar}^{Comp} standard deviations (SDs) across these areas. Culebra's K_{lidar}^{Comp} SD on October 24, 2018 was 0.0074 m^{-1}, suggesting relatively uniform water quality. In contrast, the water quality variation between the off and nearshore areas was more pronounced in Passu Keah and Marquesas Keys. Fig. 6(b) shows that Passu Keah's nearshore K_{lidar}^{Comp} exceeds 0.0888 m^{-1}, while the offshore K_{lidar}^{Comp} ranges from 0.010 to 0.062 m^{-1}. Marquesas Keys exhibited the K_{lidar}^{Comp} SD of 0.0265 m^{-1}. Analysis of Figs. 2(b) and 6(e) reveals that Marquesas Keys, with water depths mostly within 4 m, has K_{lidar} significantly influenced by underwater terrain. The seafloor materials, such as sand and gravel, stirred up by wave action likely contribute to the variability in water quality among different study areas.

To quantitatively evaluate the performance of the proposed algorithm, Fig. 7 presents the scatter plots comparing K_{lidar}^{Comp} with airborne K_{lidar}, along with the R^2, MAE, MRE, and RMSE for K_{lidar}^{Comp}. About the reference data, Passu Keah utilized a period K_{lidar} results from ALS measurements, whereas Culebra and Marquesas Keys derived their depth results from ALB data.

Overall, the scatter points are closely clustered around the 1:1 line, indicating a strong correlation between K_{lidar}^{Comp} and airborne data. The mean R^2 value of the five comparison results is 0.7947 further underscoring this strong correlation. Further analysis of the precision indices showed that the mean RMSE was 0.0060 m^{-1}, the mean MRE was 0.0549 m^{-1}, and the mean MAE was 0.0044 m^{-1}, confirming the high accuracy of the K_{lidar}^{Comp} values.

For different study areas, the R^2, RMSE, MRE, and MAE for Passu Keah reached 0.8441, 0.0077 m^{-1}, 0.0885 m^{-1}, and 0.0047 m^{-1}, respectively. In Culebra, the mean R^2, RMSE, MRE, and MAE for the two groups were 0.7430, 0.0045 m^{-1}, 0.04303 m^{-1}, and 0.0036 m^{-1}, respectively. In Marquesas

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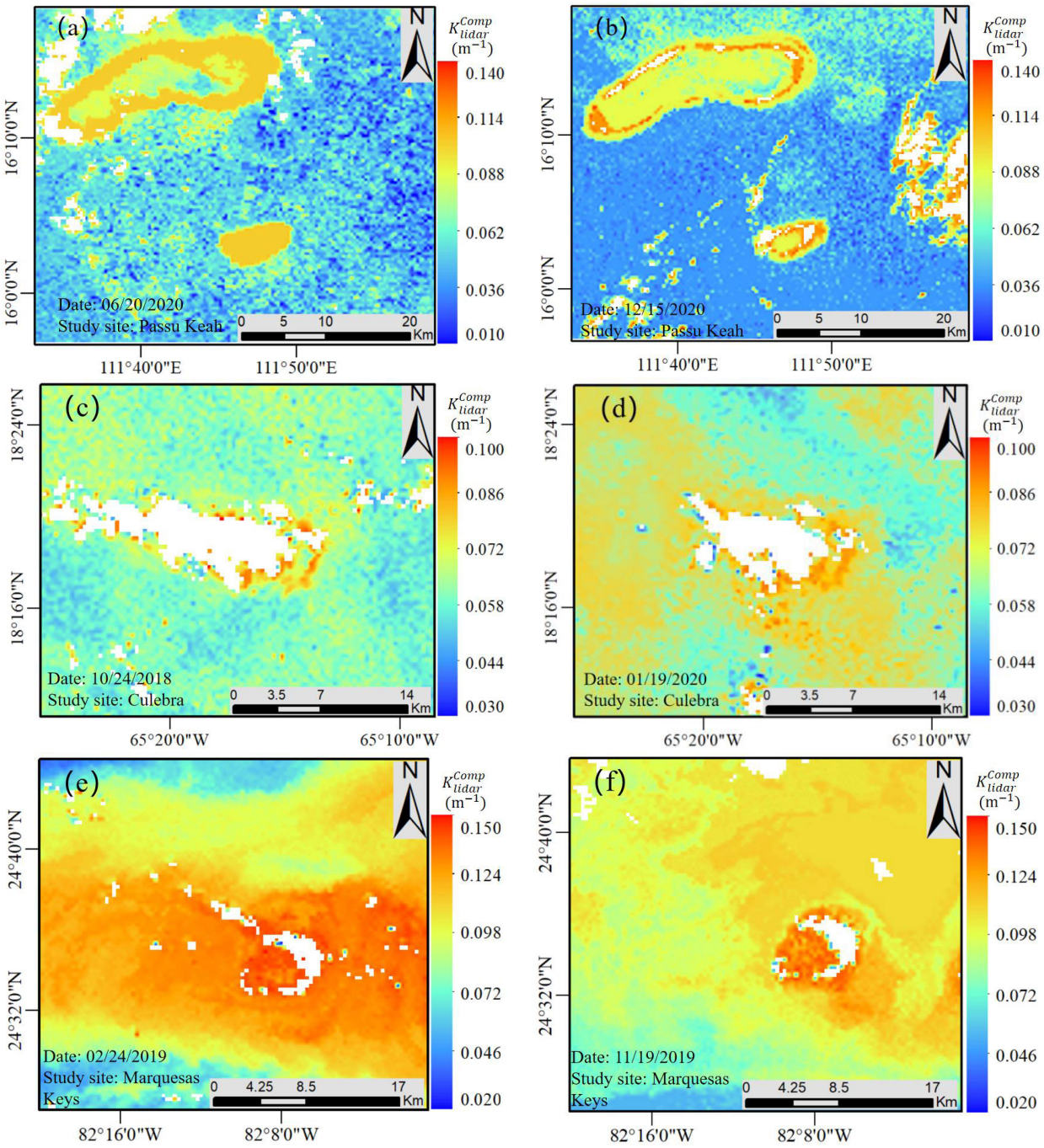


Fig. 6. $K_{\text{lidar}}^{\text{Comp}}$ results. (a) and (b) Passu Keah results, (c) and (d) Culebra results, and (e) and (f) Marquesas Keys results. Land and cloud cover are masked in white.

Keys, the mean R^2, RMSE, MRE, and MAE were 0.8216, 0.0066 m^{-1}, 0.0499 m^{-1}, and 0.0052 m^{-1}, respectively. These results demonstrate a good agreement between $K_{\text{lidar}}^{\text{Comp}}$ and the reference data, indicating that the proposed method can adapt to different seafloor terrains, sediment types, and water quality variations, and perform high-precision $K_{\text{lidar}}^{\text{Comp}}$ retrieval.

V. Discussion

A. Performance of Offshore and Nearshore K_{lidar} Results

To discuss the advantages of the fusion results over those obtained from either the offshore or nearshore regions, the performance of the K_{lidar} data retrieved from a single region is also evaluated. Fig. 8 presents the K_{lidar} retrieval results for both the offshore and nearshore regions in each of the three study areas, while Fig. 9 provides the corresponding scatter plots for comparison.

Comparing Fig. 8(a), (c), and (e) with Fig. 6(a), (c), and (e), it is observed that the $K_{\text{lidar}}^{\text{off}}$ estimates in nearshore are consistently lower. Similarly, it can be seen that $K_{\text{lidar}}^{\text{near}}$ accurately captures the variations in the turbidity of water nearshore but tends to overestimate the values in the offshore.

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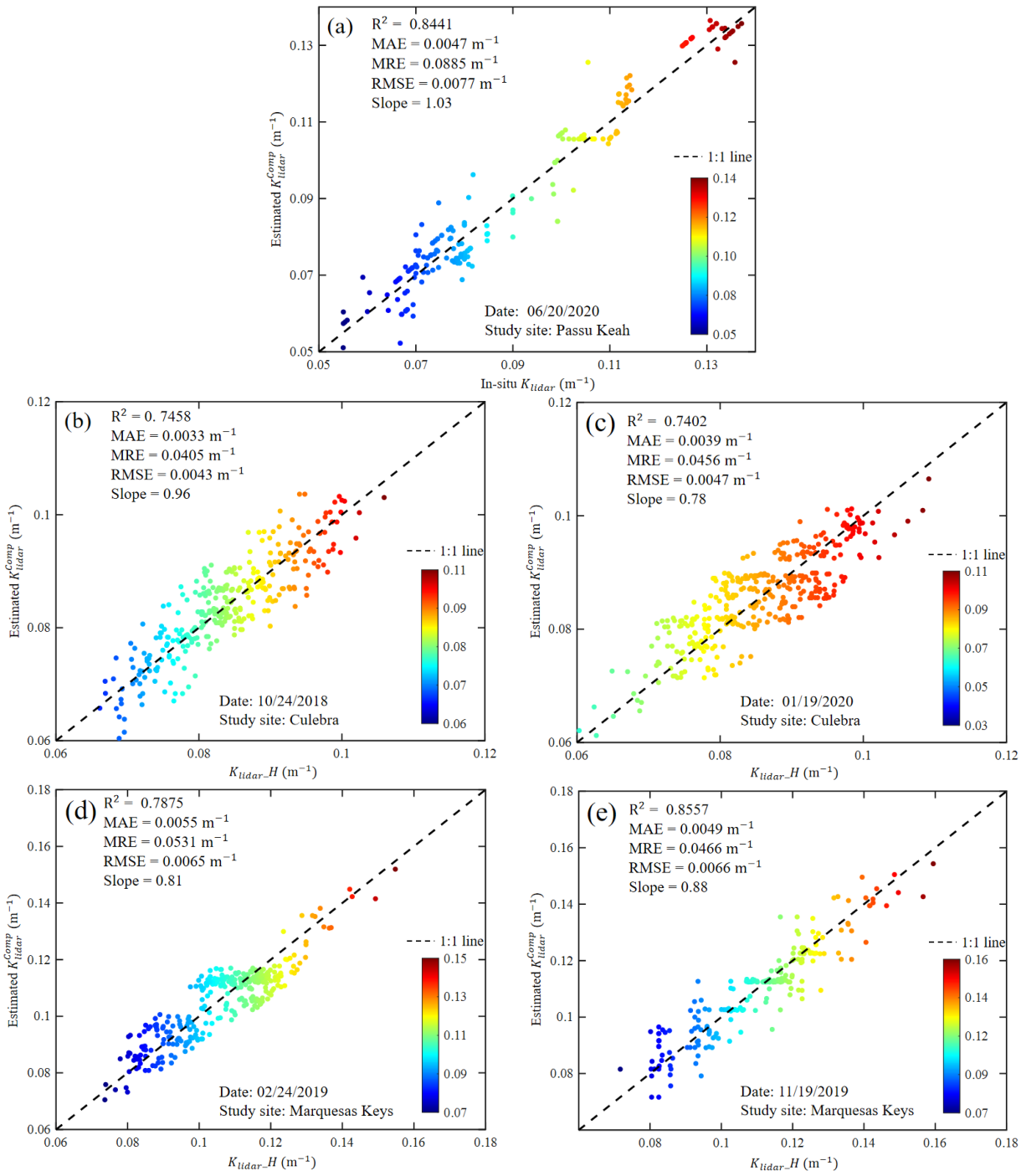


Fig. 7. Scatter plots of $K_{\text{lidar}}^{\text{Comp}}$ based on airborne data. Results for (a) Passu Keah, (b) and (c) Culebra, and (d) and (e) Marquesas Keys. The black line represents the 1:1 line and the indicators are also listed.

Table III presents the mean values of $K_{\text{lidar}}^{\text{off}}$ and $K_{\text{lidar}}^{\text{near}}$ for different study areas. When only K_{lidar}, retrieved from a single region, is used, the mean value of $K_{\text{lidar}}^{\text{off}}$ is found to be lower than that of $K_{\text{lidar}}^{\text{near}}$. This finding indicates a significant difference compared with the results presented in Table II.

Fig. 9 further validates the conclusions above. Due to the influence of underwater topography and sediment, airborne K_{lidar} values in nearshore regions are generally higher than those in offshore regions. In Fig. 9(a), (c), and (e), when the airborne K_{lidar} value is small, the scatter points remain tightly distributed around the 1:1 line, indicating high retrieval accuracy. However, with the increase of airborne K_{lidar} value, $K_{\text{lidar}}^{\text{off}}$ gradually appears to be underestimated. This indicates

TABLE III
$K_{\text{lidar}}^{\text{off}}$ AND $K_{\text{lidar}}^{\text{near}}$ MEAN VALUE

Study Site	Date	K_{lidar}^{off} mean value (m^{-1})	K_{lidar}^{near} mean value (m^{-1})
Passu Keah	06/20/2020	0.0404	0.0902
Culebra	10/24/2018	0.0427	0.0838
Marquesas Keys	02/24/2019	0.0485	0.1461

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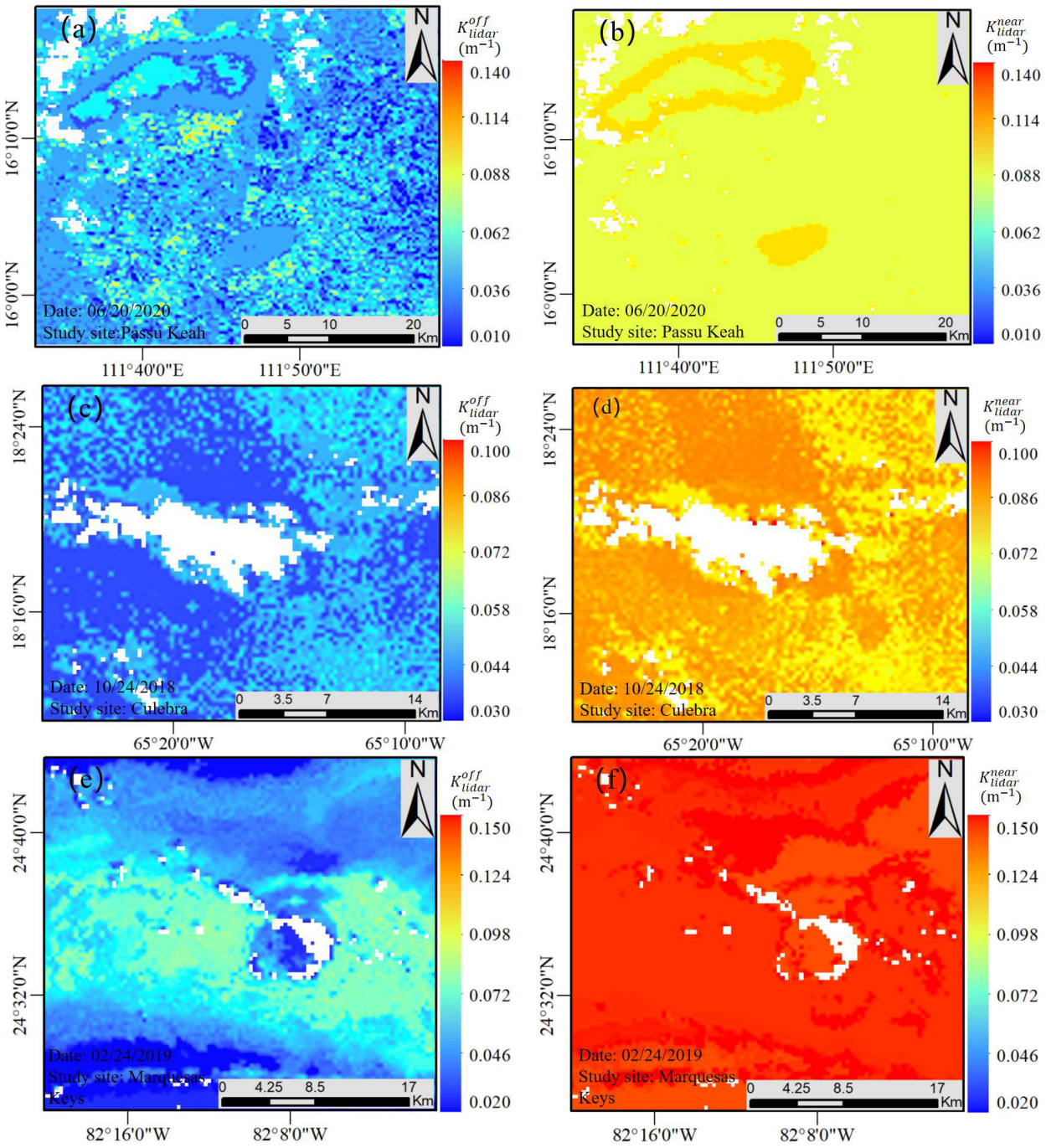


Fig. 8. $K_{\text{lidar}}^{\text{off}}$ and $K_{\text{lidar}}^{\text{near}}$ results. (a) Passu Keah $K_{\text{lidar}}^{\text{off}}$, (b) Passu Keah $K_{\text{lidar}}^{\text{near}}$, (c) Culebra $K_{\text{lidar}}^{\text{off}}$, (d) Culebra $K_{\text{lidar}}^{\text{near}}$, (e) Marquesas Keys $K_{\text{lidar}}^{\text{off}}$, and (f) Marquesas Keys $K_{\text{lidar}}^{\text{near}}$. Land and cloud cover are masked in white.

that using $K_{\text{lidar}}^{\text{off}}$ extracted from offshore water column photon signal as BO-CatBoost training samples, the model learns offshore features but fails to capture nearshore characteristics, reducing the inversion accuracy in the nearshore region. Conversely, in Fig. 9(b), (d), and (f), $K_{\text{lidar}}^{\text{near}}$, derived from seafloor photon signal retrieval in the nearshore region, is overestimated when the airborne K_{lidar} value is small. However, as the airborne K_{lidar} value increases, the scatter points gradually converge toward the 1:1 line, indicating an improvement in retrieval accuracy.

Compare with Figs. 7 and 9, the accuracy of the fusion result surpasses that of the single offshore or nearshore results obviously. On average, compared to the single result, the R^2 value of $K_{\text{lidar}}^{\text{Comp}}$ increased by 0.5195, the RMSE decreased by 0.0086 m^{-1}, the MRE and MAE decreased by 0.0856 and 0.0070 m^{-1}, respectively. These improvements verify the superiority of the fusion results in characterizing the quality of Case-I water.

B. Performance of Senitnel-3's $K_d(532)$ Product

To evaluate the superiority of the proposed method and its potential to improve the satellite detection accuracy, $K_d(490)$ obtained by Sentinel-3 is converted to $K_d(532)$ according to

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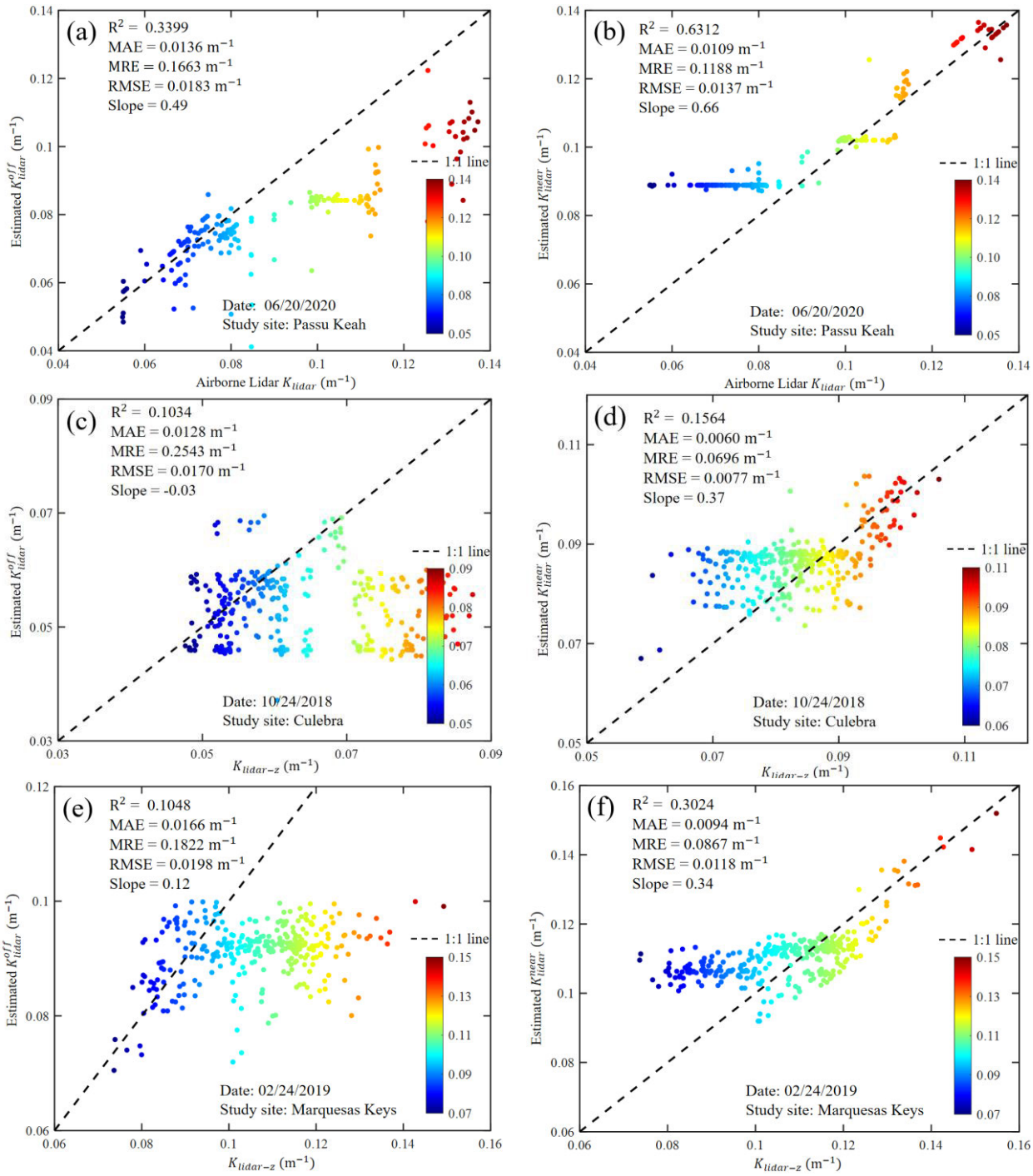


Fig. 9. Scatter plots of $K_{\text{lidar}}^{\text{off}}$ and $K_{\text{lidar}}^{\text{near}}$ based on airborne data. (a) and (b) Passu Keah, (c) and (d) Culebra, and (e) and (f) Marquesas Keys. The black line represents the 1:1 line, and the indicators are also listed.

expression (9), and compared with the airborne data. Table IV shows the mean value and SD of (532) in each study site. Fig. 10 shows the $K_d(532)$ distribution maps for each phase of the three study areas and the scatter plots for verifying accuracy based on reference data.

Overall, Fig. 10(a), (c), and (e) reveals that $K_d(532)$ exhibits significant information loss, limiting its ability to fully capture the optical characteristics of the water body in the study area. In contrast, the K_{lidar} results presented in Fig. 6(a), (c), and (e) demonstrate superior spatial coverage. From a numerical distribution perspective, K_{lidar} displays a gradual decreasing trend from nearshore to offshore regions, with rich detailed information that clearly reflects the spatial variation in the optical properties of the water body. The difference is that the $K_d(532)$ distribution map of Passu Keah shows a sharp decreasing trend with more pronounced spatial variability, failing to adequately represent the transitional characteristics of the region. In Culebra, the mean $K_d(532)$ is 0.0458 m^{-1} with a 0-m^{-1} SD, indicating nearly uniform values across the entire region, as evident in Fig. 10(c). In the Marquesas

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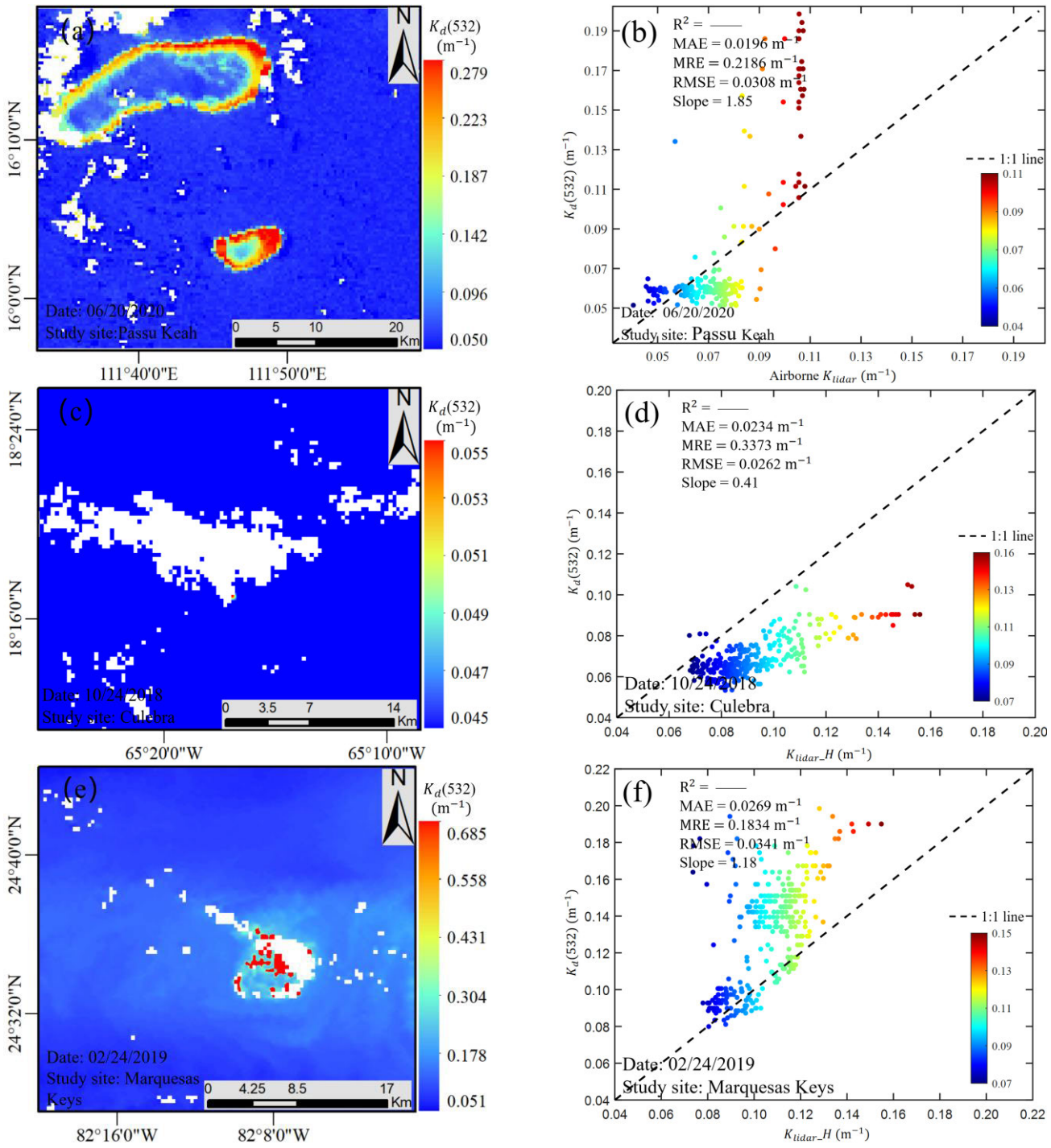


Fig. 10. $K_d(532)$ provided by Sentinel-3 and verification of its accuracy. (a) and (b) Passu Keah, (c) and (d) Culebra, and (e) and (f) Marquesas Keys. The black line represents 1:1 line, and the indicators are also listed.

Keys, $K_d(532)$ reaches exceptionally high values in extremely shallow waters, with an overall mean of 0.1150 m^{-1}, which is higher than the mean K_{lidar} value. This further highlights the limitations of $K_d(532)$ in accurately representing the apparent optical properties of the water body compared with K_{lidar}.

Fig. 10(b), (d), and (f) presents the scatter plots to show the accuracy verification of $K_d(532)$ based on the airborne LiDAR data. Overall, the correlation between $K_d(532)$ and the reference data is weak, with significant overestimation or underestimation observed in regions, where the reference values are high. In Passu Keah, the mean MAE, MRE, and RMSE of $K_d(532)$ are 0.0196, 0.2186, and 0.0308 m^{-1}, respectively. Similarly, in Culebra, these indicators are 0.0234, 0.3373, and 0.0262 m^{-1}, while in Marquesas Keys, they are 0.0269, 0.1834, and 0.0341 m^{-1}, respectively. In contrast, as shown in Fig. 7(a), (b), and (d), the accuracy and precision of K_{lidar} are significantly superior to those of $K_d(532)$. Across the three study regions, K_{lidar} demonstrates improvements in

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TABLE IV
$K_d(532)$ MEAN VALUE AND SD

Study Site	Date	Mean value (m^{-1})	SD (m^{-1})
Passu Keah	06/20/2020	0.0662	0.0260
Culebra	10/24/2018	0.0458	0.0000
Marquesas Keys	02/24/2019	0.1150	0.0368

TABLE V
OPTIMUM PARAMETER OF BO-CATBOOST

Parameter	Parameter range	Initial value	Value (offshore)	Value (nearshore)
Task type	-	-	CPU	CPU
Loss function	-	-	MAE	MAE
Iterations	10 ~ 1000	500	912	920
Depth	3 ~ 10	6	7	4
L2 regularization coefficient	10 ~ 100	15	10	3
learning rate	0.001 ~ 0.3	0.05	0.09	0.01

R^2, MAE, MRE, and RMSE with mean values of 0.7925, 0.0188 m^{-1}, 0.1861 m^{-1}, and 0.0242 m^{-1}, respectively, compared with $K_d(532)$. These statistical results underscore that K_{lidar} obtained using the proposed method exhibit significantly higher accuracy and reliability than the $K_d(532)$ provided by Sentinel-3.

C. Performance of BO-CatBoost

Although the proposed process has been proven feasible, the performance of BO and CatBoost needs to be evaluated. To this end, this study also selected XGBoost [39], a gradient boosting machine, and random forest (RF) [40] to compare their performance before and after applying BO. Previous research has suggested that CatBoost, XGBoost, and RF each have distinct characteristics. CatBoost excels at managing datasets with classification features and can effectively capture complex nonlinear relationships through gradient boosting. XGBoost is known for optimizing prediction accuracy on large datasets, while RF mitigates the risk of model overfitting and enhances model generalization by constructing multiple decision trees.

The parameters for these models encompass task type, loss function, iterations, depth, and additional specific settings. Both BO-CatBoost and BO-XGBoost include parameters, such as regularization coefficient, and learning rate. BO-RF incorporates split samples minimum and leaf samples minimum as well. Tables V–VII lists the initial settings of these parameters and document their adjustments before and after the application of BO. In both offshore and nearshore environments, BO influences the parameter combinations, which change relative to their initial settings. Moreover, the data differences between the off and nearshore regions lead to distinct model parameter requirements.

To further compare the performance of different models and verify the enhancements achieved through BO, this study compares the K_{lidar}^{comp} retrieval results obtained from these models. As depicted in Fig. 11, the K_{lidar}^{comp} results from Passu Keah are compared against the airborne K_{lidar}, with accuracy indices plotted for evaluation.

TABLE VI
OPTIMUM PARAMETER OF BO-XGBOOST

Parameter	Parameter range	Initial value	Value (offshore)	Value (nearshore)
Task type	-	-	CPU	CPU
Loss function	-	-	MAE	MAE
Iterations	10 ~ 1000	500	933	952
Depth	3 ~ 10	5	5	5
L1 regularization coefficient	0 ~ 0.01	0.005	0.002	0.009
L2 regularization coefficient	0.1 ~ 10	1	9.83	0.72
learning rate	0.001 ~ 0.3	0.1	0.3	0.15

TABLE VII
OPTIMUM PARAMETER OF BO-RF

Parameter	Parameter range	Initial value	Value (offshore)	Value (nearshore)
Task type	-	-	CPU	CPU
Loss function	-	-	MAE	MAE
Iterations	10 ~ 1000	100	10	37
Depth	3 ~ 20	10	10	3
Split samples minimum	2 ~ 20	5	2	19
Leaf samples minimum	1 ~ 20	5	1	2

Among various retrieval models evaluated, CatBoost demonstrates the best performance, with both pre- and post-BO retrieval accuracies surpassing those of XGBoost and RF. Further analysis reveals that BO-CatBoost's performance, compared with other models, shows an average increase of 0.3134 in R^2, along with an average reduction of 0.0077 m^{-1} in RMSE, 0.1922 m^{-1} in MRE, and 0.0012 m^{-1} in MAE. The results indicate that BO-CatBoost is particularly adept at capturing the complex nonlinear relationships between water quality parameters and R_{rs}, and it can most accurately represent the variations in water quality across different spatial regions. Consequently, the shallow sea K_{lidar} results retrieved using BO-CatBoost can offer researchers reliable and precise data support for their studies.

D. Analysis of Regional Feasibility

By integrating ICESat-2 and Sentinel-3 data and leveraging a Bayesian-optimized CatBoost (BO-CatBoost) model, the proposed method can achieve accurate K_{lidar} retrieval. The key innovation lies in its ability to simultaneously utilize ICESat-2's water column photons and seafloor photons, combined with Sentinel-3's multispectral images, which significantly enhances retrieval accuracy and extends its applicability. In three experimental regions, namely, Passu Keah, Culebra, and Marquesas Key, the method demonstrated exceptional performance and confirmed its high efficiency and reliability in real-world applications.

It is worth noting that the water body in these three experimental areas exhibits typical Case-I optical properties, primarily dominated by phytoplankton and their metabolites, with minimal influence from suspended solids and sediments

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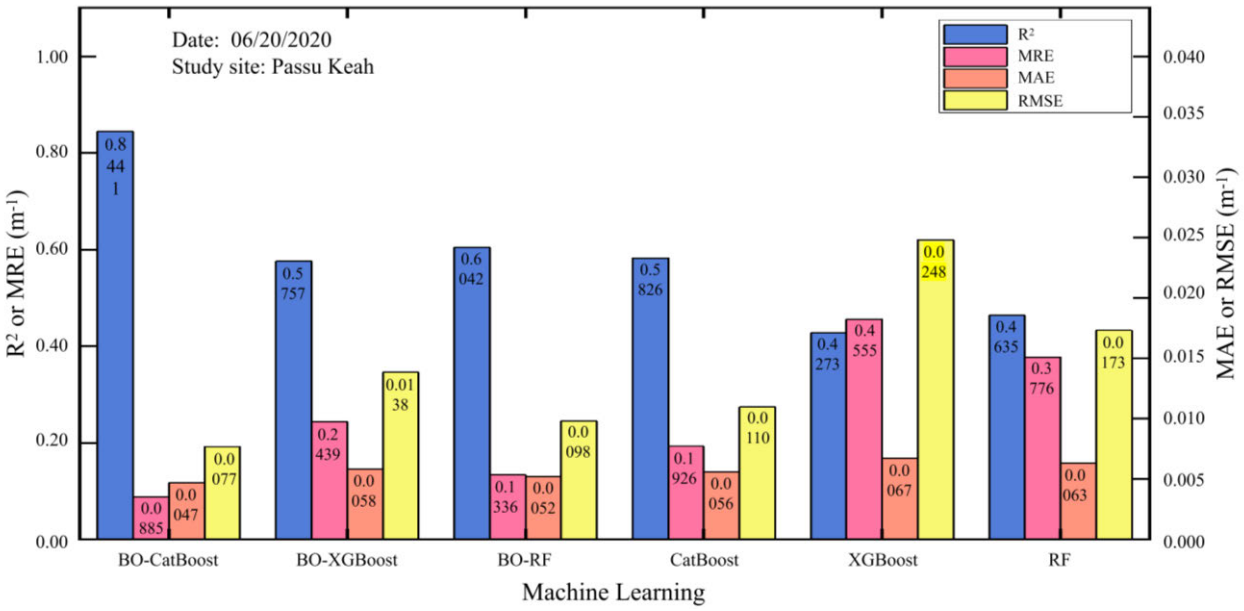


Fig. 11. Comparison of machine learning model performance. Using airborne K_{lidar} as a reference, the accuracy of various $K_{\text{lidar}}^{\text{Comp}}$ retrieval by machine learning is evaluated.

[41]. It is generally recognized that Case-I water encompass over 90% of the world's oceans and are typically found in regions, such as the Southern Ocean, the central North Pacific, and the central North Atlantic [42]. These characteristics highlight the broad applicability of the proposed method. Furthermore, with the growing availability of satellite data from ICESat-2 and Sentinel-3, the method is poised to enable large-scale K_{lidar} inversion on a global level, offering robust data support for marine resource development, water quality monitoring, and ecosystem protection.

In addition to ocean environments, Case-I water is also prevalent in inland water systems, such as lakes and reservoirs. For instance, large lakes such as Lake Fuxian [43], Lake Qinghai [44], and Lake Baikal [45] are characterized by clear water, low turbidity, and high transparency. This demonstrates that the proposed method is equally applicable to such inland water systems. Future work can focus on further optimizing the algorithm by exploring advanced machine learning models or deep learning approaches, thereby enhancing both retrieval accuracy and computational efficiency.

VI. Conclusion

The shallow sea environment is complex and changeable, making the measurement and retrieval of its apparent optical parameters through remote sensing data an effective method of monitoring water quality. This is crucial for the protection and development of coastal zones and the marine environment. This study introduces an innovative method for retrieving the LiDAR attenuation coefficient using satellite remote sensing data, integrating ICESat-2 data and Sentinel-3 images to achieve a comprehensive K_{lidar} retrieval. Initially, ICESat-2's water column and seafloor photons were utilized to calculate K_{lidar} for the offshore and nearshore regions. Then, the derived K_{lidar} from ICESat-2 and corresponding Sentinel-3 R_{rs} are combined and input into BO-CatBoost to predict K_{lidar}. Finally, the results are weighted based on the image band ratio, fusing K_{lidar} of the offshore and nearshore regions. Experimental results from Passu Keah, Culebra, and Marquesas Keys indicate that the water body's attenuation coefficient is significantly influenced by depth and seafloor sediment. The proposed method shows good performance, with an average R^2 of 0.7947, MAE of 0.0044 m^{-1}, MRE of 0.0548 m^{-1}, and RMSE of 0.0060 m^{-1} for the three study regions. The fused attenuation coefficient's accuracy surpasses that of the single-region retrieval, increasing the average R^2 by 0.5195 and reducing the average MRE, MAE, and RMSE by 0.0856, 0.0070, and 0.0086 m^{-1}, respectively. BO-CatBoost's strong ability to sense the complex nonlinear relationship between water laser attenuation and R_{rs} changes ensure accurate shallow sea K_{lidar} retrieval. Compared with $K_d(532)$ provided by Sentinel-3, K_{lidar} has better accuracy and reliability. As satellite remote sensing technology advances, this study aims to free shallow sea water quality observation from tedious field measurements, propelling related research and applications into the satellite observation era. The retrieval method combining active and passive remote sensing data can effectively generate K_{lidar} results for sensitive ecosystems such as coral reefs and seagrass beds, which are vital for marine ecosystem health, while also extending its application to clear inland lakes, offering a novel approach for monitoring inland water bodies.

It is also expected to map the spatial distribution of K_{lidar} in remote islands and reefs, providing a reliable data basis for human society's orderly development of marine resources.

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REFERENCES

[1] A. E. Morrison and T. L. Hunt, "Human impacts on the nearshore environment: An archaeological case study from Kaua'i, Hawaiian islands," *Pacific Sci.*, vol. 61, no. 3, pp. 325–345, Jul. 2007.

[2] X. Zhang et al., "Approach for estimating the vertical distribution of the diffuse attenuation coefficient in the South China Sea," *Opt. Exp.*, vol. 31, no. 26, p. 43771, Dec. 2023.

[3] A. Simon and P. Shanmugam, "Estimation of the spectral diffuse attenuation coefficient of downwelling irradiance in inland and coastal waters from hyperspectral remote sensing data: Validation with experimental data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 49, pp. 117–125, Jul. 2016.

[4] J. Yang et al., "Examining the consistency of LiDAR attenuation coefficient K_{lidar} from ICESat-2 and diffuse attenuation coefficient K_d from MODIS," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[5] C. Mitchell and A. Cunningham, "Remote sensing of spatio-temporal relationships between the partitioned absorption coefficients of phytoplankton cells and mineral particles and euphotic zone depths in a partially mixed shelf sea," *Remote Sens. Environ.*, vol. 160, pp. 193–205, Apr. 2015.

[6] S. Groom et al., "Satellite ocean colour: Current status and future perspective," *Frontiers Mar. Sci.*, vol. 6, Aug. 2019, Art. no. 485.

[7] F. Yousef, W. C. Kerfoot, C. N. Brooks, R. Shuchman, B. Sabol, and M. Graves, "Using LiDAR to reconstruct the history of a coastal environment influenced by legacy mining," *J. Great Lakes Res.*, vol. 39, pp. 205–216, 2013.

[8] A. Baltodano, A. Agramont, K. Lekarkar, E. Spyrakos, I. Reusen, and A. van Griensven, "Exploring global remote sensing products for water quality assessment: Lake Nicaragua case study," *Remote Sens. Appl., Soc. Environ.*, vol. 36, Nov. 2024, Art. no. 101331.

[9] Y. Zhang, C. Giardino, and L. Li, "Water optics and water colour remote sensing," *Remote Sens.*, vol. 9, no. 8, p. 818, Aug. 2017.

[10] Y. Sun, D. Wang, L. Li, R. Ning, S. Yu, and N. Gao, "Application of remote sensing technology in water quality monitoring: From traditional approaches to artificial intelligence," *Water Res.*, vol. 267, Dec. 2024, Art. no. 122546.

[11] Z. Lee, K. L. Carder, and R. A. Arnone, "Deriving inherent optical properties from water color: A multiband quasi-analytical algorithm for optically deep waters," *Appl. Opt.*, vol. 41, no. 27, p. 5755, Sep. 2002.

[12] G. Zheng, D. Stramski, and R. A. Reynolds, "Evaluation of the quasi-analytical algorithm for estimating the inherent optical properties of seawater from ocean color: Comparison of Arctic and lower-latitude waters," *Remote Sens. Environ.*, vol. 155, pp. 194–209, Dec. 2014.

[13] Y. Wang, F. Shen, L. Sokoletsky, and X. Sun, "Validation and calibration of QAA algorithm for CDOM absorption retrieval in the Changjiang (Yangtze) estuarine and coastal waters," *Remote Sens.*, vol. 9, no. 11, p. 1192, Nov. 2017.

[14] S. Kratzer, C. Brockmann, and G. Moore, "Using MERIS full resolution data to monitor coastal waters—A case study from Himmerfjärden, a fjord-like bay in the northwestern Baltic sea," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2284–2300, May 2008.

[15] A. Morel, Y. Huot, B. Gentili, P. J. Werdell, S. B. Hooker, and B. A. Franz, "Examining the consistency of products derived from various ocean color sensors in open ocean (case 1) waters in the perspective of a multi-sensor approach," *Remote Sens. Environ.*, vol. 111, no. 1, pp. 69–88, Nov. 2007.

[16] M. Nazeer, M. Bilal, M. Alsahli, M. Shahzad, and A. Waqas, "Evaluation of empirical and machine learning algorithms for estimation of coastal water quality parameters," *ISPRS Int. J. Geo-Inf.*, vol. 6, no. 11, p. 360, Nov. 2017.

[17] C. A. Hostetler, M. J. Behrenfeld, Y. Hu, J. W. Hair, and J. A. Schulien, "Spaceborne LiDAR in the study of marine systems," *Annu. Rev. Marine Sci.*, vol. 10, pp. 121–147, Dec. 2018.

[18] K. Richter, D. Mader, P. Westfeld, and H.-G. Maas, "Refined geometric modeling of laser pulse propagation in airborne LiDAR bathymetry: Verbesserte geometrische modellierung der laserpulsausbreitung in der laserbathymetrie," *PFG J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 89, no. 2, pp. 121–137, Apr. 2021.

[19] L. Magruder, T. Neumann, and N. Kurtz, "ICESat-2 early mission synopsis and observatory performance," *Earth Space Sci.*, vol. 8, no. 5, May 2021, Art. no. e2020EA001555.

[20] Y. Ma et al., "Satellite-derived bathymetry using the ICESat-2 LiDAR and Sentinel-2 imagery datasets," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112047.

[21] X. Lu, Y. Hu, and Y. Yang, "Ocean subsurface study from ICESat-2 mission," in *Proc. Photon. Electromagn. Res. Symp.-Fall*, Dec. 2019, pp. 910–918.

[22] X. Lu, Y. Hu, Y. Yang, P. Bontempi, A. Omar, and R. Baize, "Antarctic spring ice-edge blooms observed from space by ICESat-2," *Remote Sens. Environ.*, vol. 245, Aug. 2020, Art. no. 111827.

[23] R. H. Watkins, M. J. Sayers, R. A. Shuchman, and K. R. Bosse, "Validation of ICESat-2 derived data products on freshwater lakes: Bathymetry, diffuse attenuation coefficient for downwelling irradiance (K_d), and particulate backscatter coefficient (b_{bp})," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[24] X. Zhang, Y. Ma, Z. Li, and J. Zhang, "Satellite derived bathymetry based on ICESat-2 diffuse attenuation signal without prior information," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 102993.

[25] L. Chen, G. Zhang, S. Xing, Z. Wang, R. Kong, and Q. Xu, "Satellite retrieval of LiDAR attenuation coefficient from ICESat-2 and Sentinel-3 based on machine learning: Inland waters," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[26] J. Zhu et al., "Bathymetry retrieval without in-situ depth using an ICESat-2 assisted dual-band model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17739–17752, 2024.

[27] Y. Wang et al., "Satellite retrieval of benthic reflectance by combining LiDAR and passive high-resolution imagery: Case-I water," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112955.

[28] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Proc. 32nd Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 31, Montreal, QC, Canada, Dec. 2018, pp. 6639–6649.

[29] J. Li et al., "Range difference between shallow and deep channels of airborne bathymetry LiDAR with segmented field-of-view receivers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jun 2022, Art. no. 5703616.

[30] B. Billard, R. H. Abbot, and M. F. Penny, "Airborne estimation of sea turbidity parameters from the WRELADS laser airborne depth sounder," *Appl. Opt.*, vol. 25, no. 13, p. 2080, Jul. 1986.

[31] Z. Leng, J. Zhang, Y. Ma, J. Zhang, and H. Zhu, "A novel bathymetry signal photon extraction algorithm for photon-counting LiDAR based on adaptive elliptical neighborhood," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 115, Dec. 2022, Art. no. 103080.

[32] G. Zhang, S. Xing, Q. Xu, P. Li, and D. Wang, "A pre-pruning quadtree isolation method with changing threshold for ICESat-2 bathymetric photon extraction," *Remote Sens.*, vol. 15, no. 6, p. 1629, Mar. 2023.

[33] L. Chen, S. Xing, G. Zhang, S. Guo, and M. Gao, "Refraction correction based on ATL03 photon parameter tracking for improving ICESat-2 bathymetry accuracy," *Remote Sens.*, vol. 16, no. 1, p. 84, Dec. 2023.

[34] C. Parrish, L. Magruder, A. Neuenschwander, N. Forfinski-Sarkozi, M. Alonzo, and M. Jasinski, "Validation of ICESat-2 ATLAS bathymetry and analysis of ATLAS's bathymetric mapping performance," *Remote Sens.*, vol. 11, no. 14, p. 1634, Jul. 2019.

[35] R. P. Stumpf, K. Holderied, and M. Sinclair, "Determination of water depth with high-resolution satellite imagery over variable bottom types," *Limnol. Oceanogr.*, vol. 48, pp. 547–556, Jan. 2003.

[36] A. Morel and S. Maritorena, "Bio-optical properties of oceanic waters: A reappraisal," *J. Geophys. Res., Oceans*, vol. 106, no. C4, pp. 7163–7180, Apr. 2001.

[37] J. Li et al., "Adaptive bathymetry estimation for shallow coastal waters using planet dove satellites," *Remote Sens. Environ.*, vol. 232, Oct. 2019, Art. no. 111302.

[38] M. Wang, S. Son, and L. W. Harding, "Retrieval of diffuse attenuation coefficient in the Chesapeake bay and turbid ocean regions for satellite ocean color applications," *J. Geophys. Res., Oceans*, vol. 114, Oct 2009, Art. no. 111302.

[39] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 785–794.

[40] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, Oct. 2001.

[41] A. Morel and L. Prieur, "Analysis of variations in ocean color," *Limnol. Oceanogr.*, vol. 22, no. 4, pp. 709–722, Jul. 1977.

[42] IOCCG, *Remote Sensing of Ocean Colour in Coastal, and Other Optically-Complex*, S. W. Sathyendranath, Eds., Canada: Reports of the International Ocean-Colour Coordinating Group, No. 3, IOCCG, 2000.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:11:49 UTC from IEEE Xplore. Restrictions apply.

[43] Q. Han, J. Sun, Q. Chou, T. Cao, W. Li, and Y. Cao, "To be or not to be: The fate of submerged macrophyte biodiversity in the Plateau lakes after restoration for the last decade," *J. Environ. Manage.*, vol. 370, Nov. 2024, Art. no. 123085.
[44] X. Ban et al., "Estimation of phytoplankton primary productivity in Qinghai lake using ocean color satellite data: Seasonal and interannual variations," *Water*, vol. 16, no. 10, p. 1433, May 2024.
[45] C. Sansom, "Preserving lake Baikal," *Frontiers Ecol. Environ.*, vol. 2, no. 10, p. 508, Dec. 2004.



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